

PROJECT: AHB-APB BRIDGE

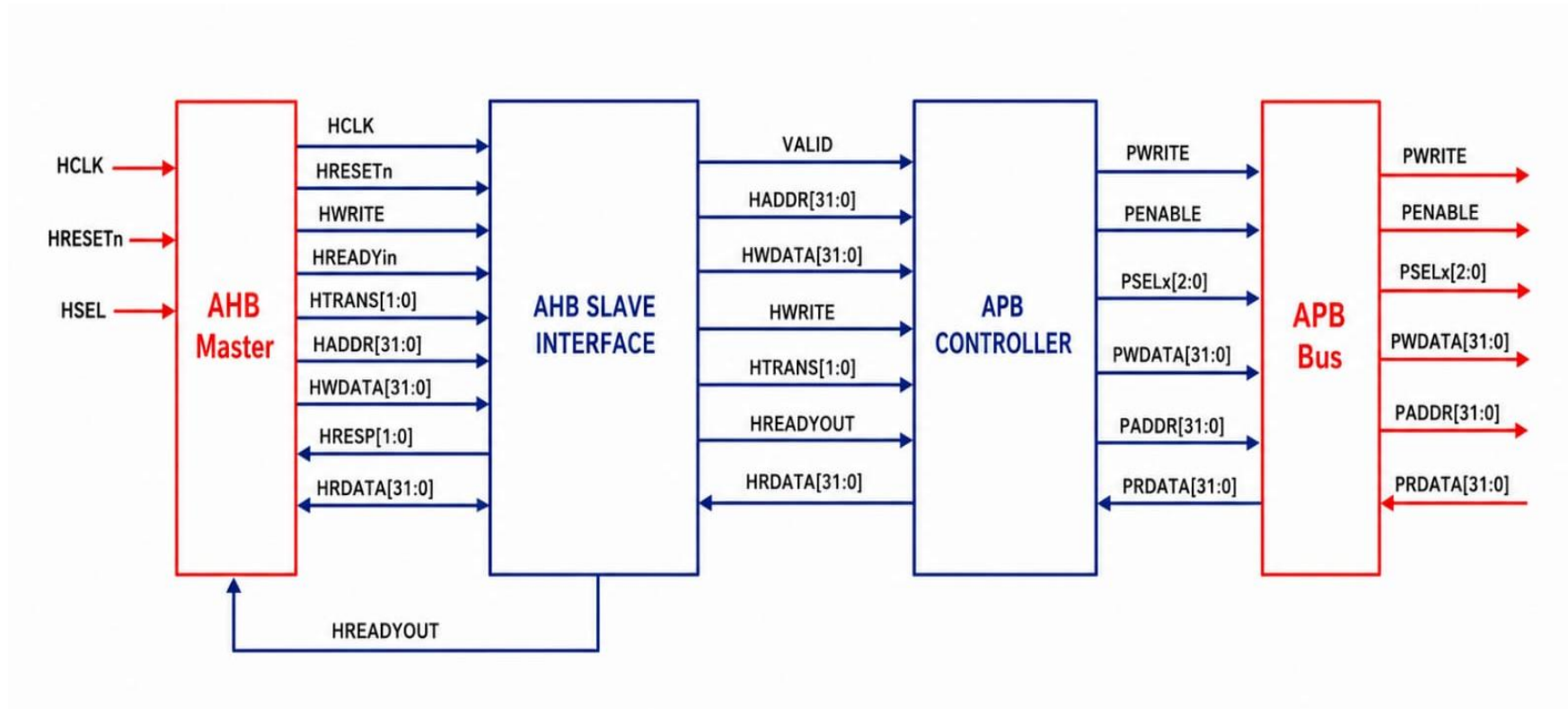
From,
PRAKRUTHI BY
(Batch :- DI 65)

To,
Maven Silicon
Bengaluru, India

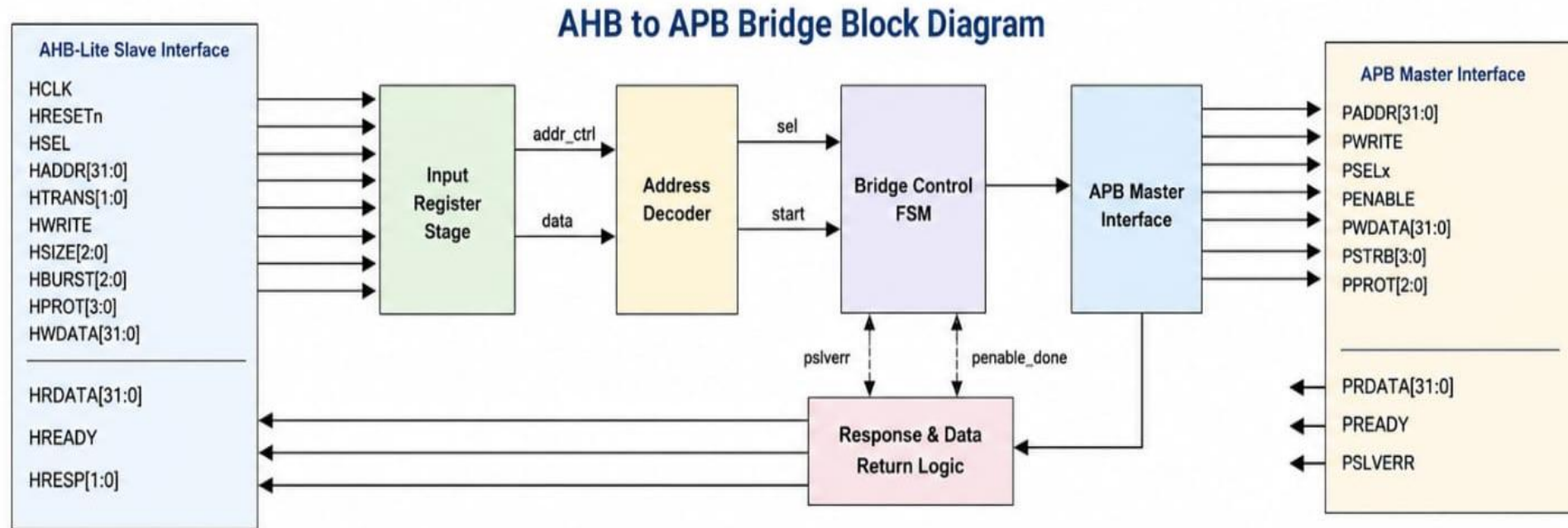
Report content

- Architecture
- Block Diagram
- Each block with waveforms with Finite state machine(FSM) and Simulation methodology
- Synthesis
- Conclusion on the project

ARCHITECTURE



BLOCK DIAGRAM



EACH BLOCK WITH WAVEFORM

AHB INTERFACE

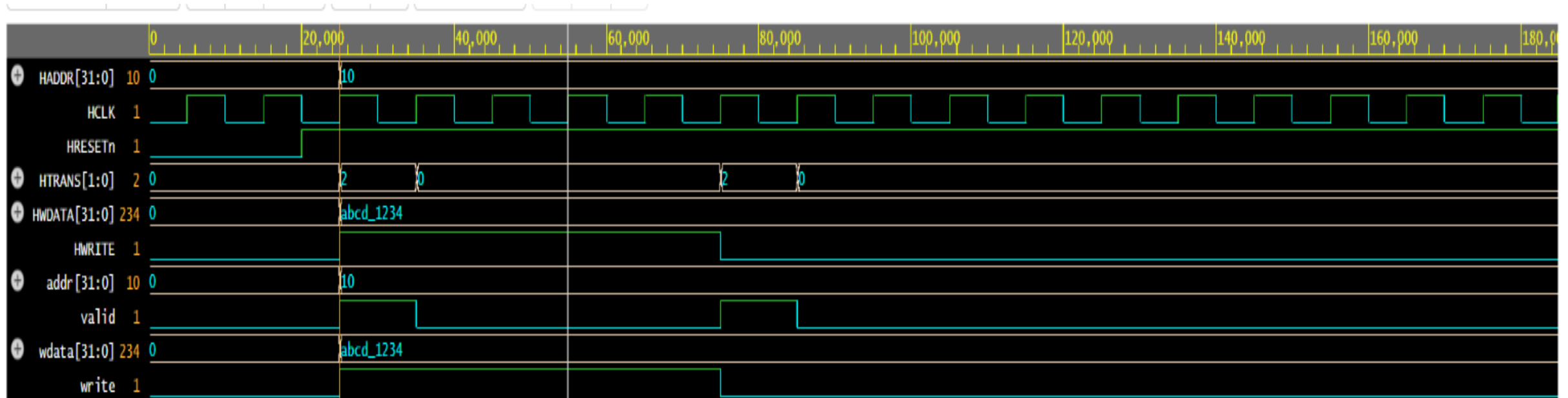
- The Advanced High-performance Bus interface is the communication interface used in high-speed systems inside an SoC.
- It is part of the AMBA protocol family developed by Arm Holdings.
- AHB is mainly used for:
- CPU communication, Memory transfers , DMA communication , High-speed peripherals
- **Function**
- The AHB interface receives address, control, and data information from the master and forwards the transaction details to the bridge controller for protocol conversion.



Important AHB Signals

Signal	Description
HCLK	System clock for AHB bus
HRESETn	Active-low reset signal
HADDR	Address bus carrying transfer address
HWRITE	Indicates read (0) or write (1) operation
HWDATA	Data written from master to slave
HRDATA	Data read from slave to master
HREADY	Indicates completion of transfer
HTRANS	Transfer type information

AHB INTERFACE WAVEFORM



APB CONTROLLER

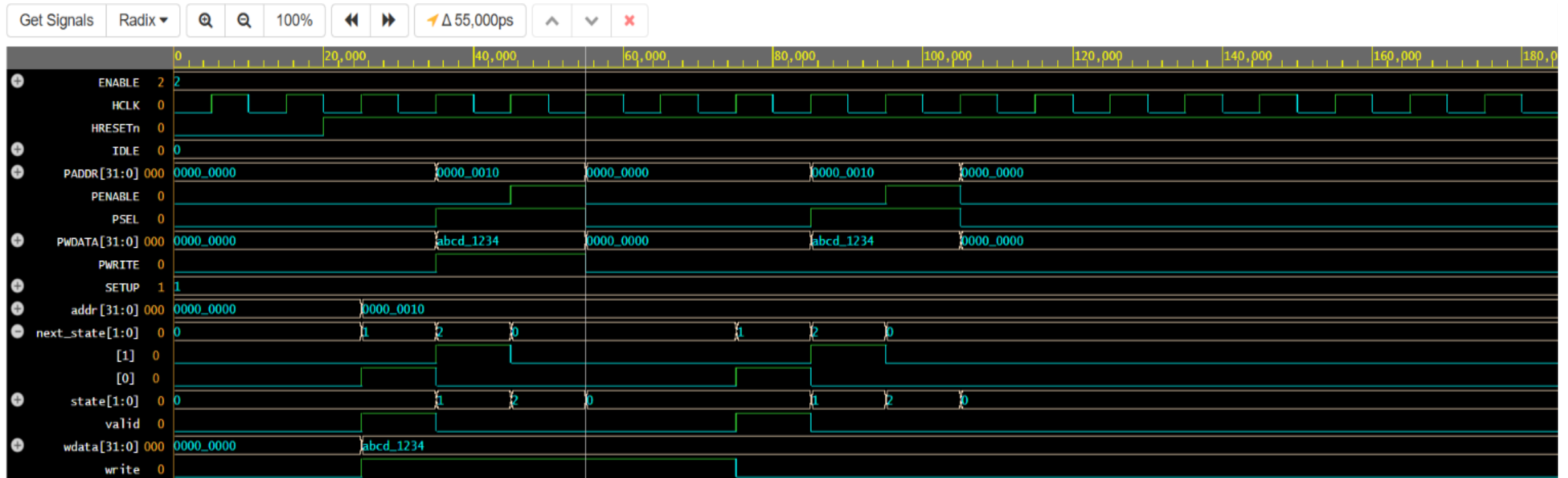
- The Advanced Peripheral Bus (APB) controller is the logic used to communicate with low-speed peripheral devices in an AMBA-based SoC.
- It is mainly designed for:
 - Low power
 - Simple communication
 - Low complexity peripherals
- **Function**
 - The APB interface communicates with low-power peripheral devices and performs read/write operations using a simple two-phase transfer mechanism.



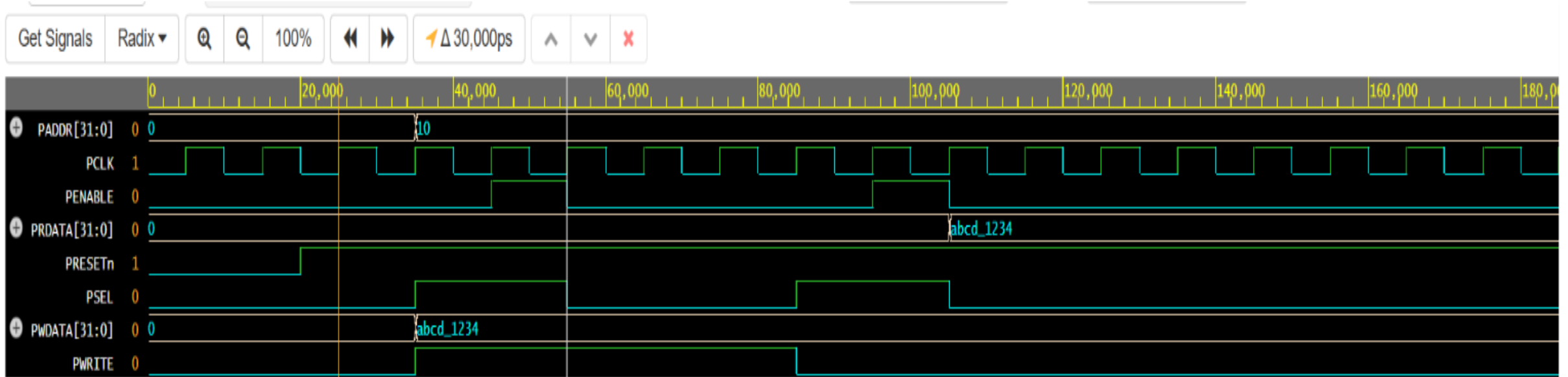
Important APB Signals

Signal	Description
PCLK	APB clock signal
PRESETn	Active-low reset
PADDR	Peripheral address bus
PWRITE	Read/Write control
PWDATA	Write data bus
PRDATA	Read data bus
PSEL	Peripheral select signal
PENABLE	Enables APB transfer
PREADY	Indicates transfer completion

APB CONTROLLER WAVEFORM

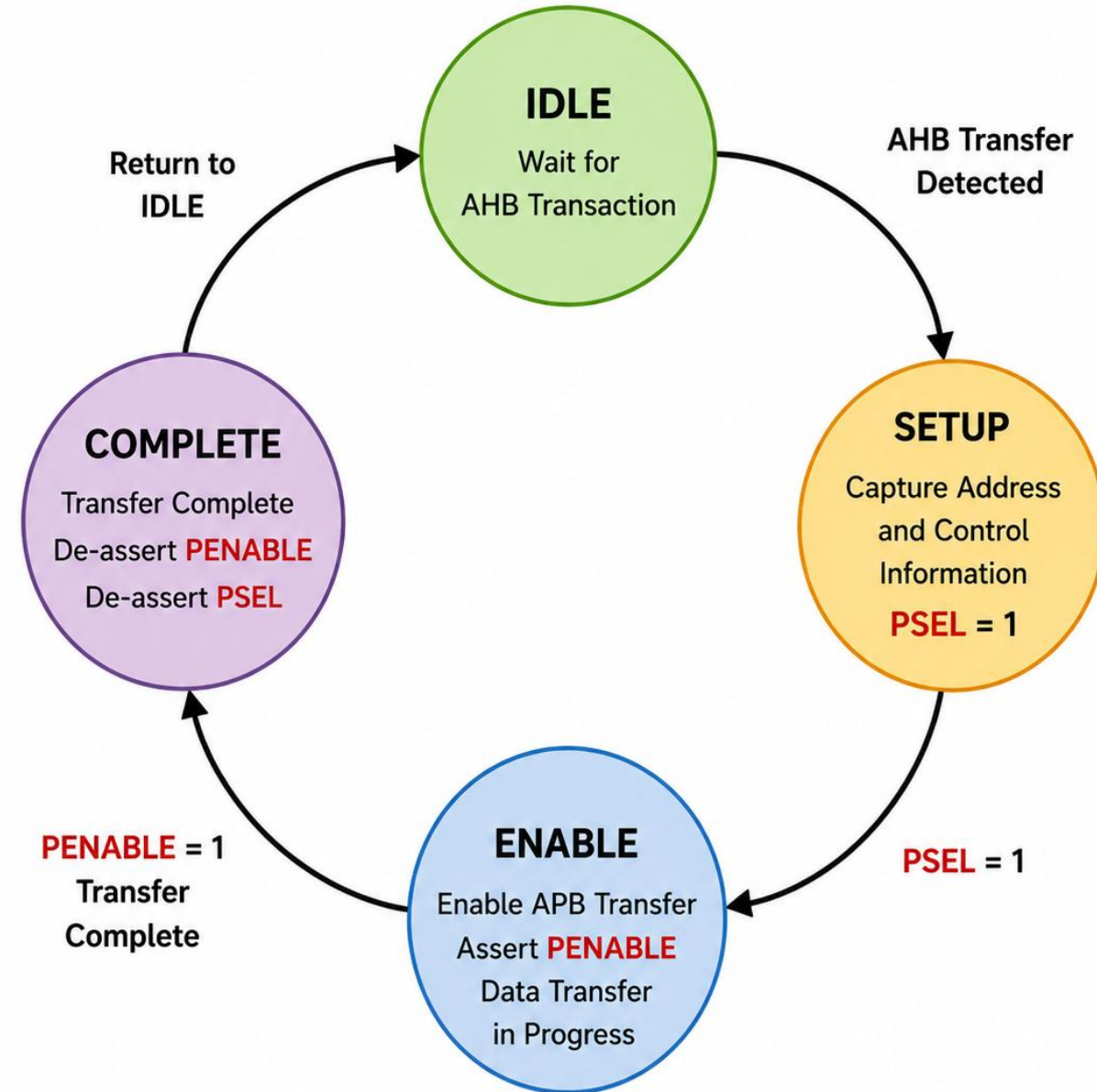


APB SLAVE WAVEFORM



FINITE STATE MACHINE (FSM)

- **FSM of APB Controller**
- **IDLE State:** Waits for a valid AHB transaction.
- **SETUP State:** Captures address and control information and asserts PSEL.
- **ENABLE State:** Asserts PENABLE and performs APB data transfer.
- **COMPLETE State:** Completes the transaction and returns to the IDLE state.



Simulation Methodology

Tools Used

ModelSim / QuestaSim Simulator

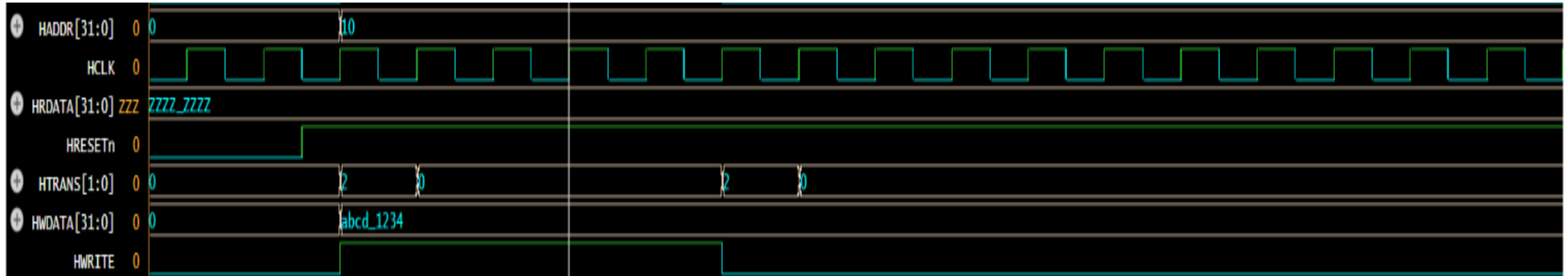
Verification Process

1. The AHB-to-APB Bridge was verified using a Verilog testbench developed to generate AHB read and write transactions.
2. Functional verification was performed by monitoring address, data, and control signals during protocol conversion.
3. Waveforms of the AHB Interface, APB Controller, APB Slave, and DUT were analyzed to validate transfer operations.
4. Simulation results were compared with the expected behavior to ensure correct communication between AHB and APB buses.

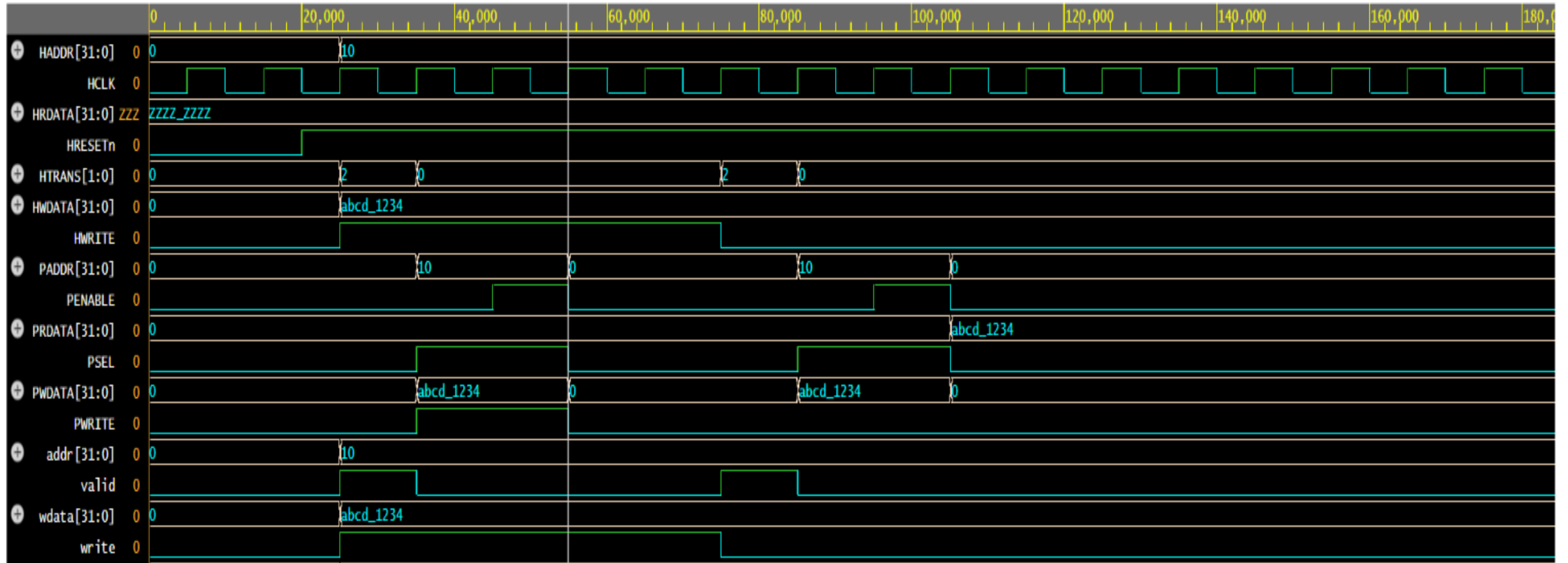
Result

Successful protocol conversion and reliable read/write data transfer were achieved. All modules operated according to AMBA protocol requirements

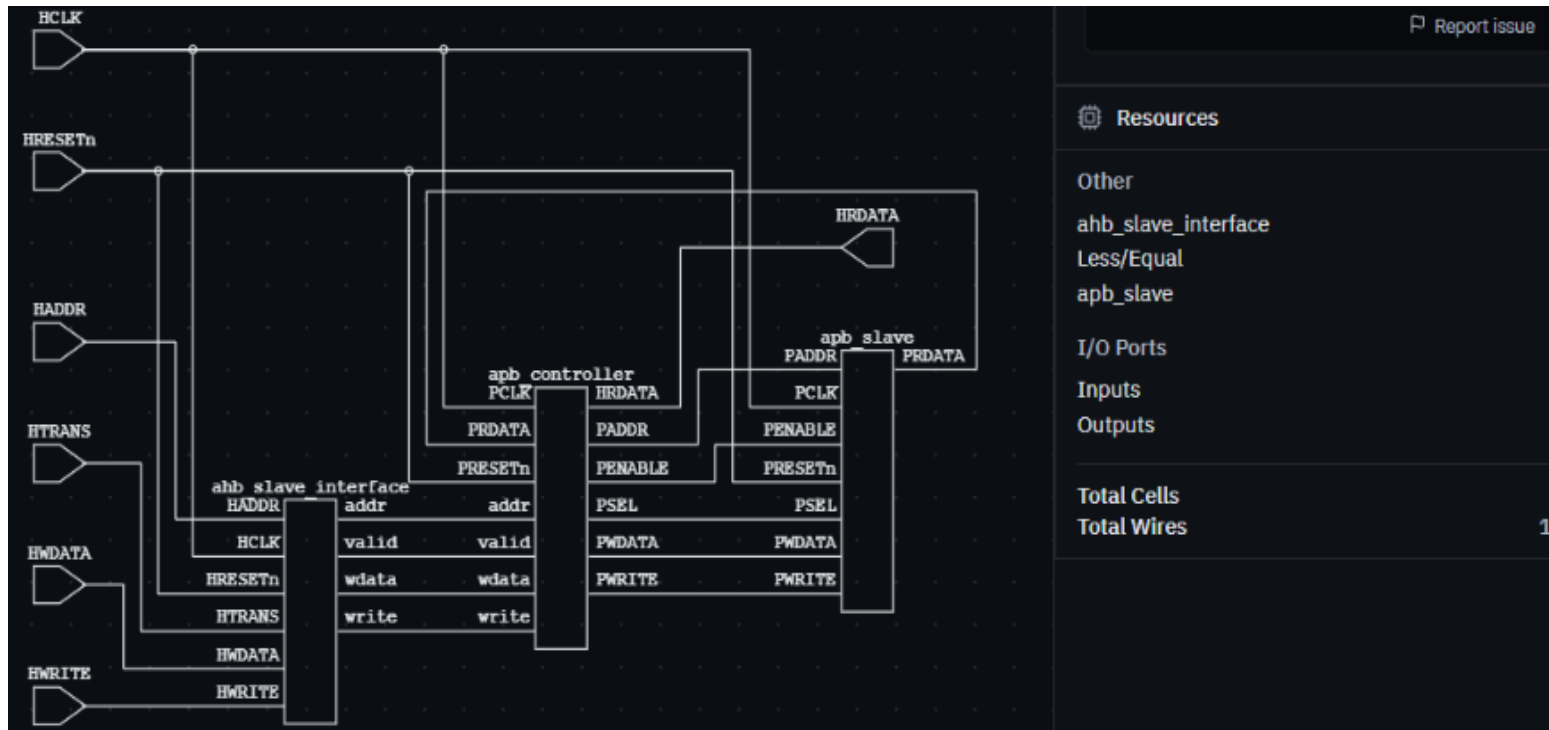
TESTBENCH WAVEFORM



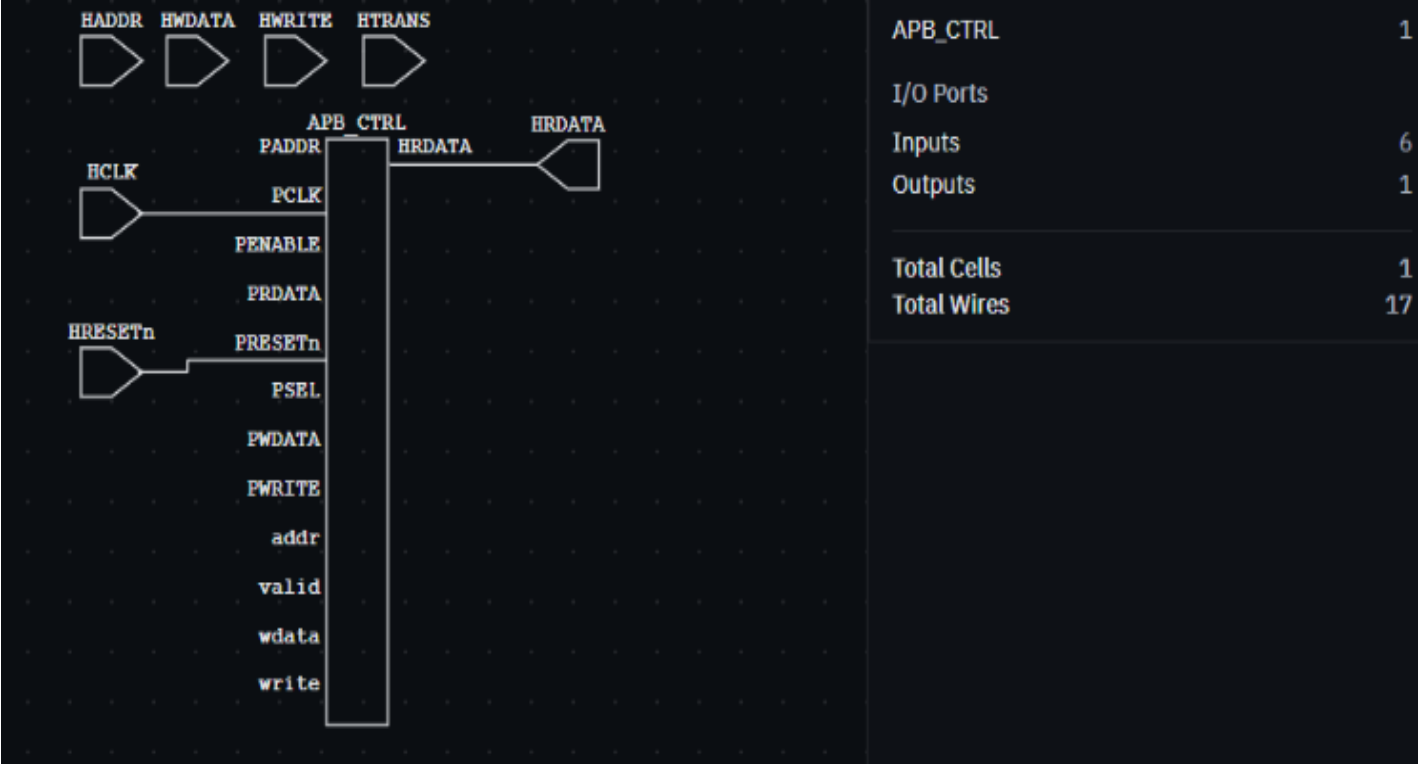
DESIGN UNDER TEST WAVEFORM



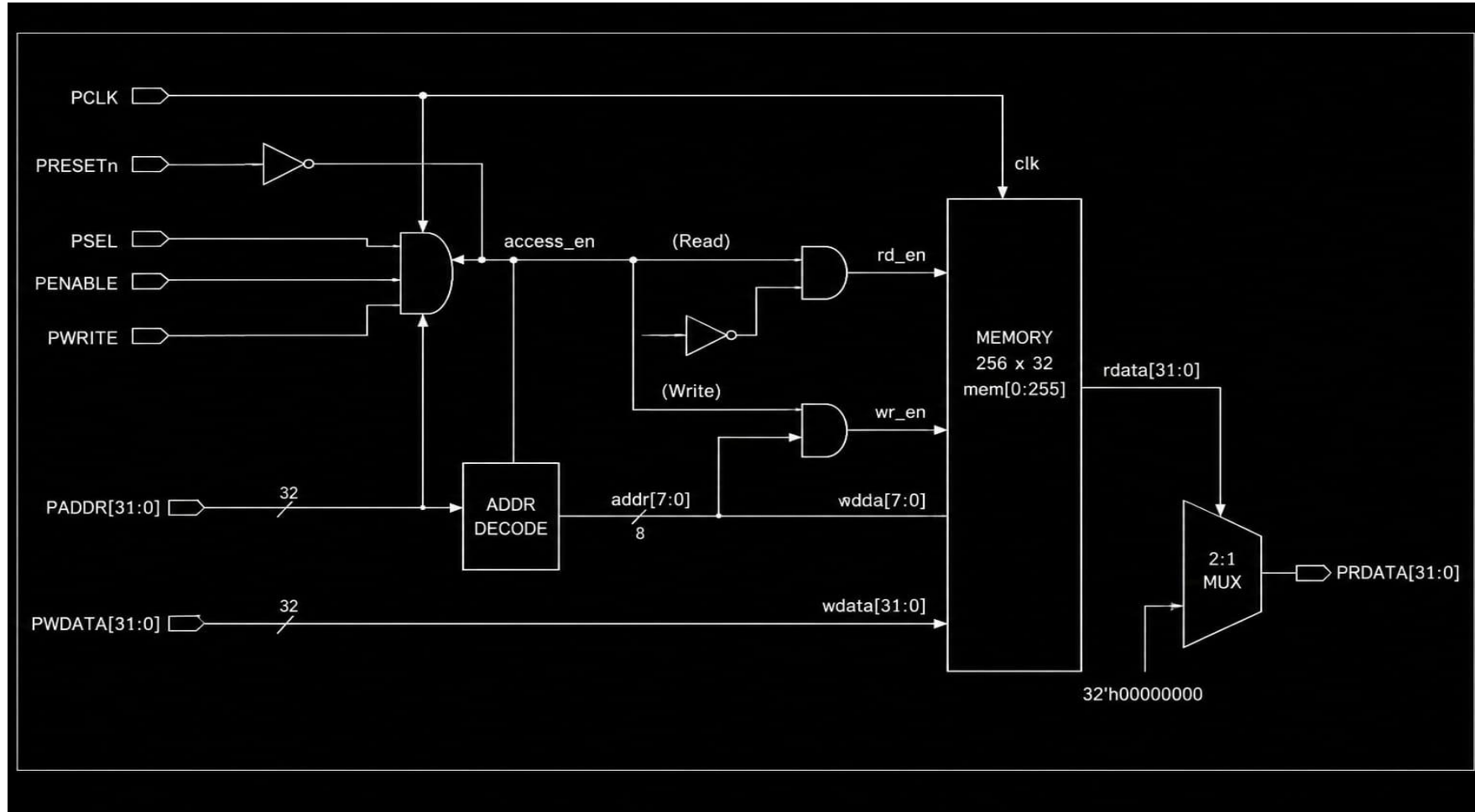
SYNTHESIS: AHB APB BRIDGE



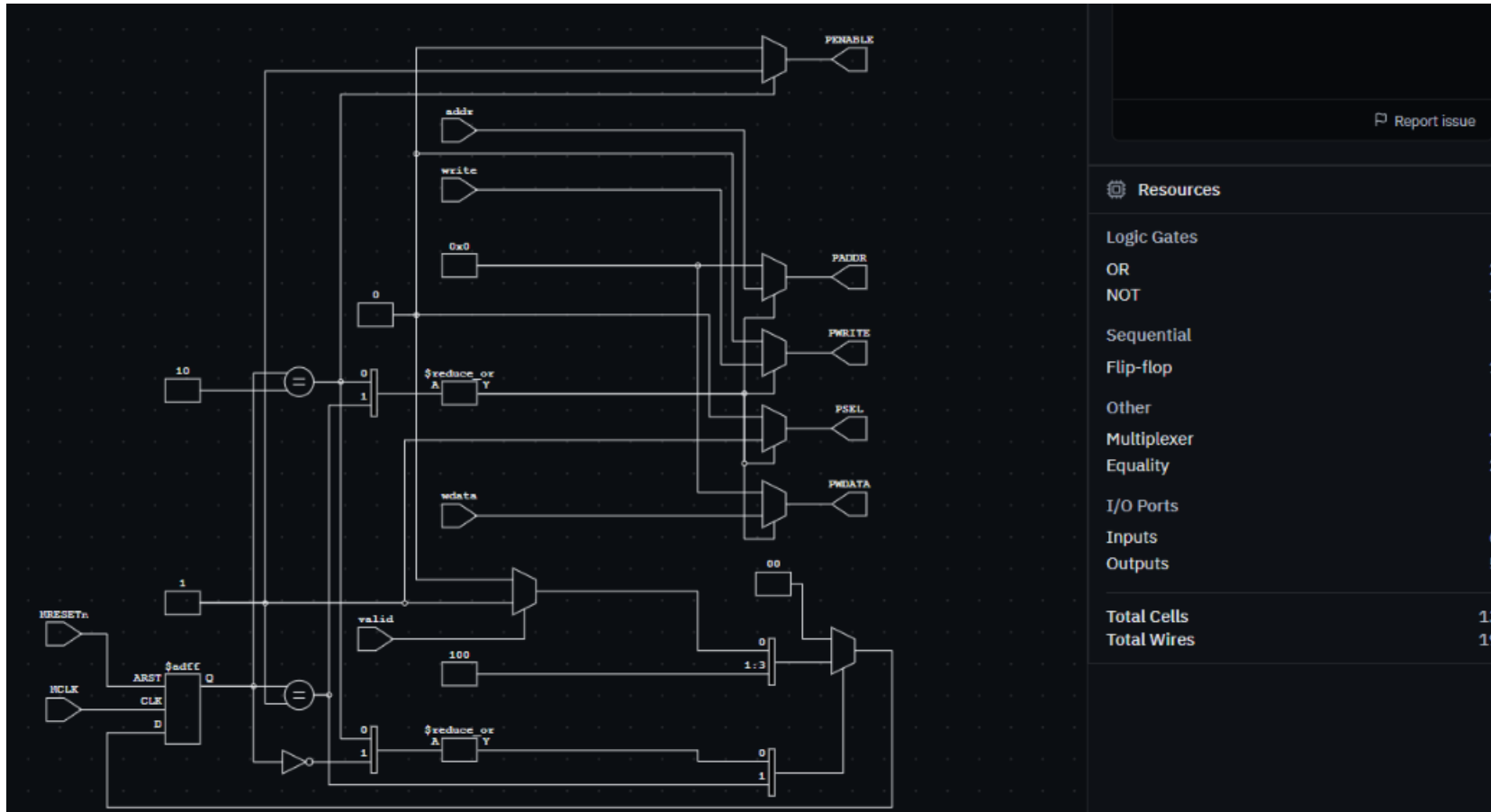
APB CONTROLLER

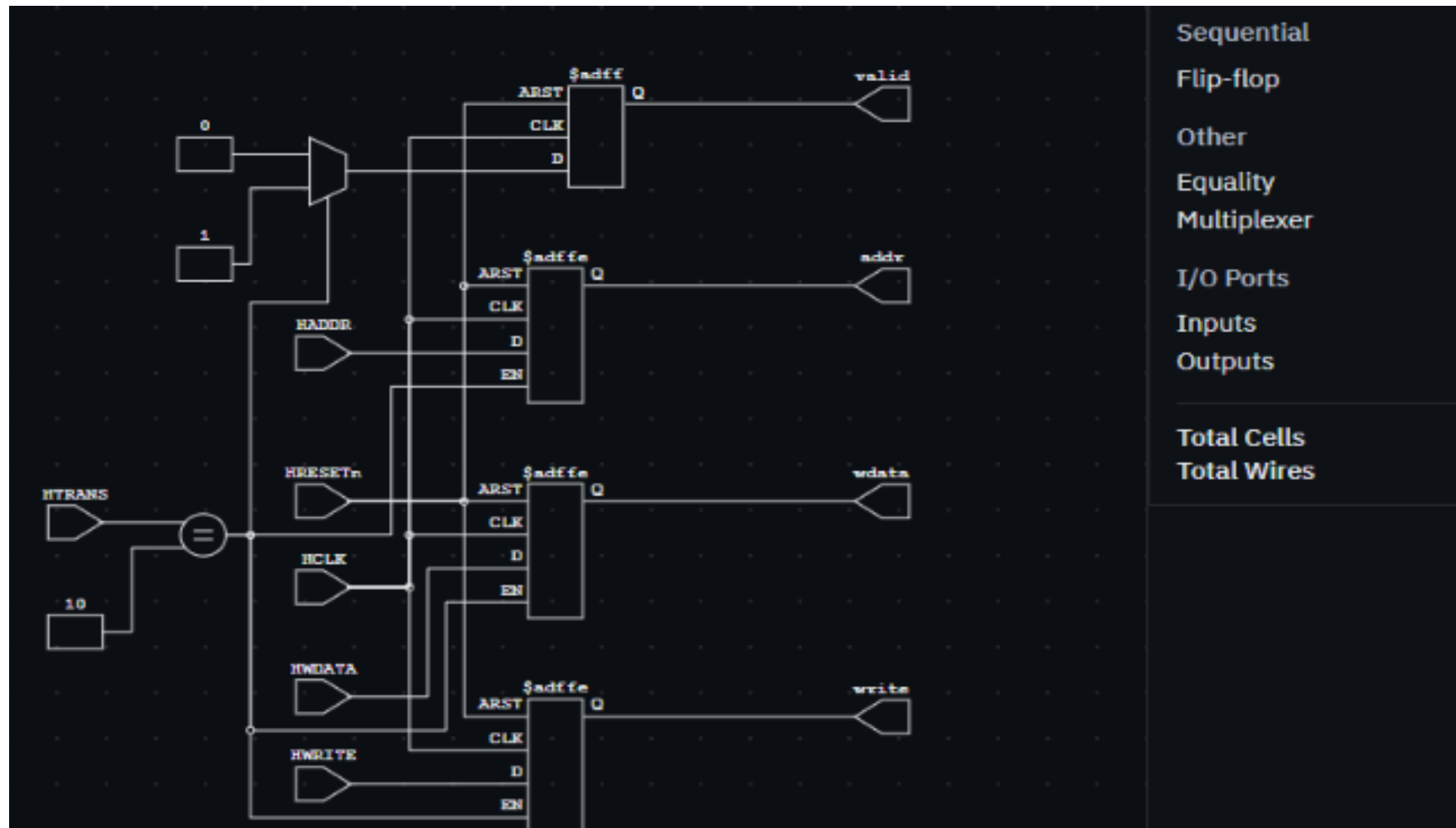


APB SLAVE



AHB SLAVE INTERFACE





CONCLUSION

The AHB to APB Bridge project successfully implements communication between the high-speed AHB bus and the low-power APB bus using Verilog HDL. The bridge efficiently converts AHB transactions into APB-compatible transfers while supporting proper read/write operations, address decoding, and control signal generation. The design ensures reliable data transfer and smooth communication between different AMBA protocols, making it suitable for SoC, FPGA, and ASIC-based applications. This project also provides a strong understanding of bus protocols, FSM-based control design, and RTL implementation.